FOR QC MORPHISMS, CLOSEDNESS IS THE SAME AS SPECIALIZATION LIFTING

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3916-3933, following the statement at 3908-3914.

Editorial note. The step 'the restriction of f to Z is still quasi-compact' (~3924-3925) is uncited but standard (a closed immersion is quasi-compact and qc is preserved under composition).

Allowed standard prerequisites (editorial).

- Closed maps lift specializations. Topology, Lemma topology-lemma-closed-open-map-specialization (cross-chapter, cited at ~3918). (1) direction.
- Reduced induced scheme structure. definition-reduced-induced-scheme (same chapter, cited at ~3923). Views the closed subset Z as a scheme.
- Specialization lifting and composition. Topology, Lemma topology-lemma-lift-specialization-composition (cross-chapter, cited at ~3927). Specializations lift along Z -> S.
- Images of qc maps stable under specializations. Topology, Lemma topology-lemma-lift-specializations-images (cross-chapter, cited at ~3930). f(X) is specialization-stable.
- Specialization-stable image of a qc map is closed. lemma-image-quasi-compact-closed (same chapter, cited at ~3932). Concluding engine.

Source reference. Original definition 01J4: sources/stacks-schemes/pinned-original-source.tex, source lines 2456–2456. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05JL: sources/stacks-schemes/pinned-original-source.tex, source lines 3883–3883. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0063: focused-seeds/geometry/S9-schemes-quasi-compact-closed/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0066: focused-seeds/geometry/S9-schemes-quasi-compact-closed/ref-tags.txt, source lines 6–6. Navigation pointer only: the

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counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0064: focused-seeds/geometry/S9-schemes-quasi-compact-closed/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0065: focused-seeds/geometry/S9-schemes-quasi-compact-closed/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a quasi-compact morphism of schemes. Then f is closed if and only if specializations lift along f, see Topology, Definition 0063.*

Proof. According to Topology, Lemma 0066 if f is closed then specializations lift along f. Conversely, suppose that specializations lift along f. Let $Z \subset X$ be a closed subset. We may think of Z as a scheme with the reduced induced scheme structure, see Definition 01J4. Since $Z \subset X$ is closed the restriction of f to Z is still quasi-compact. Moreover specializations lift along $Z \to S$ as well, see Topology, Lemma 0064. Hence it suffices to prove $f(X)$ is closed if specializations lift along f. In particular $f(X)$ is stable under specializations, see Topology, Lemma 0065. Thus $f(X)$ is closed by Lemma 05JL. □